Advanced Drone Technology

INDIA SPACE LAB

Project Report

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Abstract

Aircraft conceptual design is a critical phase in aerospace engineering where the overall configuration and geometry of an aircraft are defined before performing detailed aerodynamic, structural, and performance analyses. In modern aerospace engineering, computational design tools are widely used to develop accurate and efficient aircraft models. One such tool is OpenVSP (Open Vehicle Sketch Pad), an open-source parametric aircraft modeling software developed by NASA.

The purpose of this project is to design and visualize a simplified aircraft model inspired by the Cessna 210 configuration using OpenVSP. The modeling process includes creating major aircraft components such as the fuselage, main wing, horizontal stabilizer, vertical stabilizer, and propeller system. The aircraft geometry was developed using parametric design principles, allowing dimensions and positions of various components to be adjusted dynamically.

The project focuses on understanding the structural configuration of a conventional propeller-driven aircraft and the process of constructing aircraft geometry using computer-aided design tools. Through this project, the modeling workflow used in conceptual aircraft design was explored and implemented. The final aircraft model represents a complete aircraft configuration and demonstrates the integration of multiple aerodynamic components within a single design framework

Introduction

Aircraft design is a multidisciplinary field that integrates various engineering domains such as aerodynamics, propulsion systems, structural mechanics, avionics, and flight control systems. The design of an aircraft begins with the conceptual design phase, where engineers define the overall geometry and configuration of the aircraft.

In the conceptual design stage, the goal is to determine the basic layout of the aircraft, including the fuselage shape, wing configuration, tail arrangement, and propulsion system. This stage focuses on establishing the geometric parameters that influence aircraft performance and stability.

With advancements in aerospace engineering, computational tools have become essential for developing aircraft geometries efficiently. Computer-aided design software allows engineers to rapidly prototype aircraft models and evaluate their aerodynamic characteristics before proceeding to detailed engineering design.

OpenVSP (Open Vehicle Sketch Pad) is one of the most widely used conceptual aircraft design tools. Developed by NASA, OpenVSP allows engineers to create parametric aircraft models using adjustable geometric parameters such as span, chord length, sweep angle, and taper ratio.

In this project, OpenVSP was used to design a simplified aircraft model inspired by the Cessna 210, a single-engine high-performance aircraft commonly used in general aviation. The project demonstrates how aircraft geometry can be constructed step-by-step by combining various parametric components.

Background of Aircraft Geometry Design

Aircraft geometry plays a significant role in determining the aerodynamic performance, stability, and structural integrity of an aircraft. The external shape of an aircraft influences parameters such as lift generation, drag characteristics, and flight stability.

The major components of aircraft geometry include:

Fuselage

The fuselage is the central body of the aircraft that houses the cockpit, passengers, cargo, and various onboard systems. The fuselage shape affects aerodynamic drag and structural efficiency.

Wing

The wing is the primary lift-generating component of the aircraft. The wing geometry determines the aircraft's lift characteristics, stall behavior, and overall aerodynamic efficiency.

Horizontal Stabilizer

The horizontal stabilizer provides longitudinal stability to the aircraft and helps control pitch motion during flight.

Vertical Stabilizer

The vertical stabilizer ensures directional stability and prevents unwanted yaw motion.

Propulsion System

The propulsion system generates thrust to propel the aircraft forward. In conventional light aircraft, propulsion is typically provided by a propeller-driven engine.

Understanding the interaction between these components is essential for designing efficient aircraft configurations.

Objective of the Project

The primary objective of this project is to design and visualize a simplified aircraft geometry using OpenVSP software while understanding the fundamental principles of aircraft configuration design.

The specific objectives include**:**

• To understand the interface and capabilities of OpenVSP software.

• To create fuselage geometry using cross-section editing techniques.

• To design the main wing structure using parametric wing parameters.

• To construct horizontal and vertical stabilizers for aircraft stability.

• To add and position a propeller geometry to represent the propulsion system.

• To visualize the aircraft configuration from different orthographic and isometricviews.

• To understand how individual aircraft components are integrated to form a complete aircraft structure

Software Used

OpenVSP (Open Vehicle Sketch Pad):

OpenVSP is an open-source aircraft geometry modeling tool developed by NASA for conceptual aircraft design. The software enables engineers to rapidly create aircraft models using parametric inputs.

Key features of OpenVSP include:

- Parametric aircraft geometry modeling
- Cross-section editing tools for fuselage design
- Wing geometry configuration with adjustable aerodynamic parameters
- Visualization of aircraft models from multiple views
- Integration with aerodynamic analysis tools

OpenVSP is widely used by aerospace engineers, researchers, and students for conceptual aircraft design and aerodynamic modeling.

Reference Aircraft Configuration

The aircraft modeled in this project is inspired by the Cessna 210, a widely known single-engine light aircraft used in general aviation.

The Cessna 210 features a high-wing configuration, which provides good stability and improved visibility for the pilot. It is commonly used for private transport, aerial photography, and survey operations.

Typical features of the reference aircraft include :

- Single-engine propeller propulsion system
- High-wing monoplane configuration
- Conventional tail design
- Lightweight fuselage structure
- Retractable landing gear (not modeled in this project)

Reference drawings of the aircraft provide approximate dimensions such as wingspan, fuselage length, and cabin width. These measurements help estimate the proportions of the aircraft geometry during modeling.

Methodology

The aircraft geometry was developed using a systematic modeling approach in OpenVSP. Each component of the aircraft was created separately and then assembled into a complete aircraft configuration.

The methodology followed in this project includes the following steps:

Step 1: Setting Measurement Units

The first step in the modeling process was to configure the measurement units within OpenVSP. Since the reference aircraft dimensions were provided in feet, the measurement units were set accordingly to maintain consistency with the reference drawings.

Procedure:

1. Open OpenVSP software.
2. Navigate to Model → Measure.
3. Set the length unit to Feet.

This ensures that all aircraft dimensions correspond accurately to the reference measurements used for the Cessna-210 based model.

Step 2: Fuselage Creation

The fuselage geometry was created using the Fuselage geometry tool in OpenVSP. The fuselage shape was constructed by modifying cross-sections using the XSec Editor, allowing the model to match the general shape of the reference aircraft.

Steps performed:

1. Select Add → Fuselage.
2. Adjust the fuselage length to approximately 24.5 ft based on the reference drawing.
3. Use the XSec Editor to modify multiple fuselage cross-sections.
4. Adjust the following parameters:

- Nose curvature for aerodynamic shaping

- Cabin diameter approximately 5.17 ft
- Maximum fuselage width around 3.85 ft
- Tail cone taper to smoothly reduce fuselage diameter
- Cross-section alignment for smooth geometry transition

Multiple cross-sections were edited to achieve a smooth fuselage profile that closely resembles the reference aircraft geometry.

Step 3: Main Wing Design

The main wing geometry was created using the Wing geometry tool in OpenVSP.

Steps performed:

1. Select Add → Wing.
2. Configure the following wing parameters:

- Wing span: approximately 36.7 ft
- Root chord: approximately 5 ft
- Tip chord: approximately 2.5 ft
- Taper ratio: approximately 0.5
- Sweep angle: approximately 2–3 degrees
- Dihedral angle: approximately 3 degrees

3. Position the wing relative to the fuselage using X and Z coordinates.

The wing was positioned slightly above the fuselage to replicate the high-wing configuration of the reference aircraft. This configuration improves stability and visibility in light aircraft designs.

Step 4: Horizontal Tail Creation

The horizontal stabilizer was created by adding another wing geometry with smaller dimensions compared to the main wing.

Steps performed:

1. Add a new Wing Geometry.
2. Adjust the stabilizer parameters:

- Span: approximately 12.9 ft
- Root chord: approximately 3 ft
- Tip chord: approximately 1.5 ft
- Taper ratio: approximately 0.5

3. Position the geometry at the rear section of the fuselage.
4. Adjust alignment using X, Y, and Z coordinates.

The horizontal stabilizer provides pitch stability and helps maintain aircraft longitudinal balance during flight.

Step 5: Vertical Tail Creation

The vertical stabilizer was created by adding another wing geometry and rotating it by 90° to orient it vertically.

Steps performed:

1. Add a new Wing Geometry.
2. Rotate the geometry by 90° to make it vertical.
3. Adjust the following parameters:

- Height: approximately 5–6 ft
- Root chord: approximately 3 ft
- Sweep angle: approximately 15 degrees

4. Position the stabilizer at the rear fuselage section.

The vertical stabilizer provides directional stability and assists in maintaining aircraft yaw control.

Step 6: Propeller Creation

The propulsion system was represented by adding a propeller geometry at the nose of the aircraft.

Steps performed:

1. Select Add → Propeller.
2. Configure the following parameters:

- Propeller diameter: approximately 6 ft
- Number of blades: 2 blades

3. Position the propeller at the nose section of the fuselage.

The propeller represents the engine-driven propulsion system, which generates thrust required for aircraft forward motion.

Results And Model Visualization

After completing the geometry modeling process, the aircraft model was visualized using multiple viewing perspectives within OpenVSP.

The following views were captured:

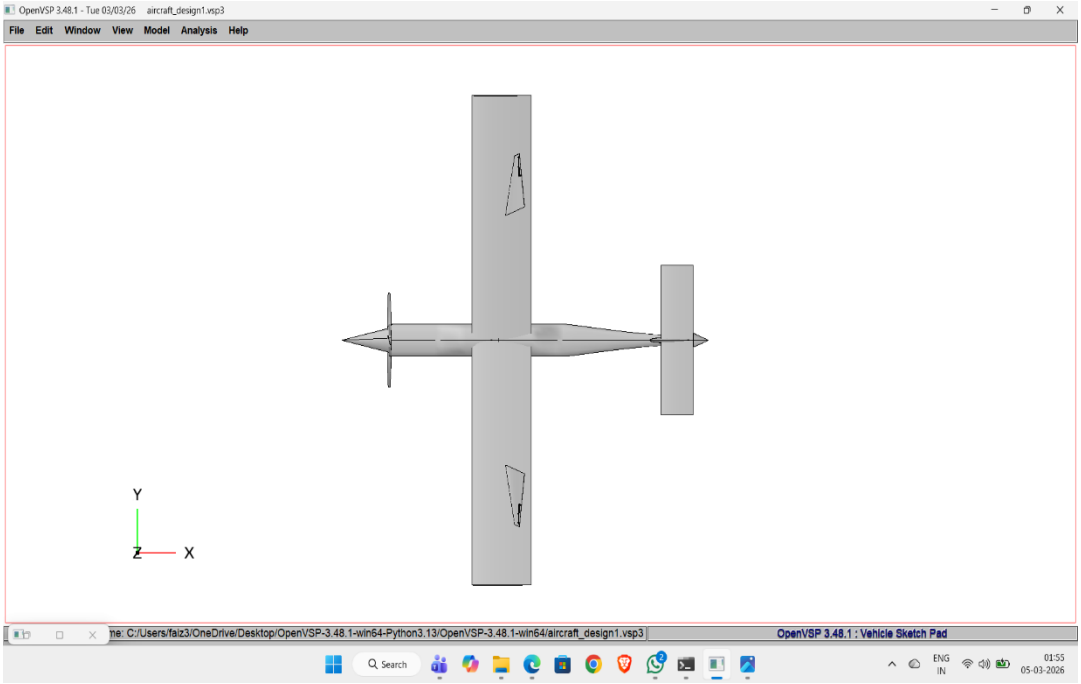

FIG 1 : Top View

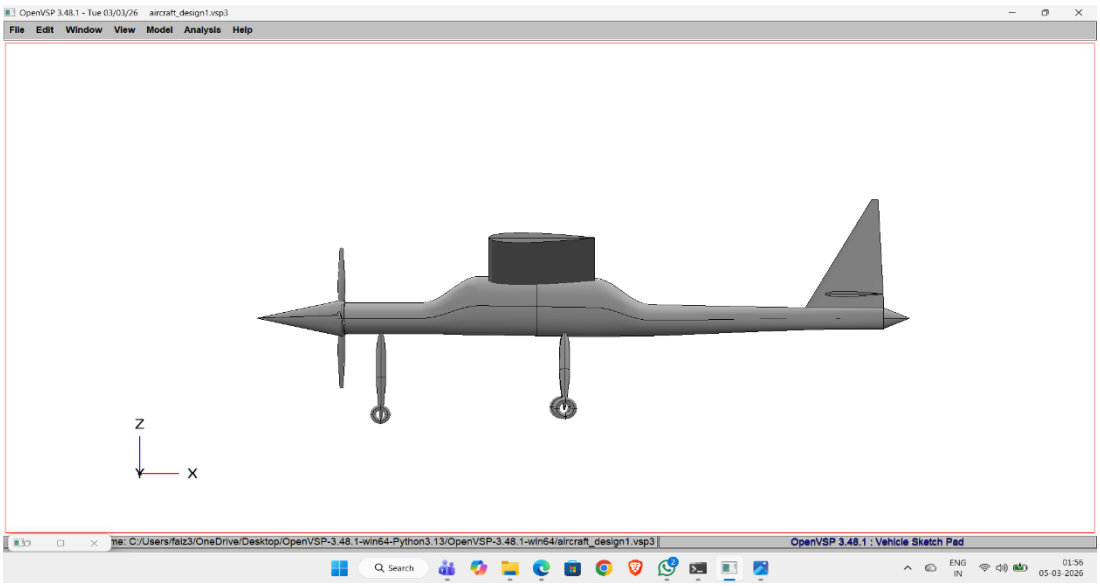

FIG 2 : Side View

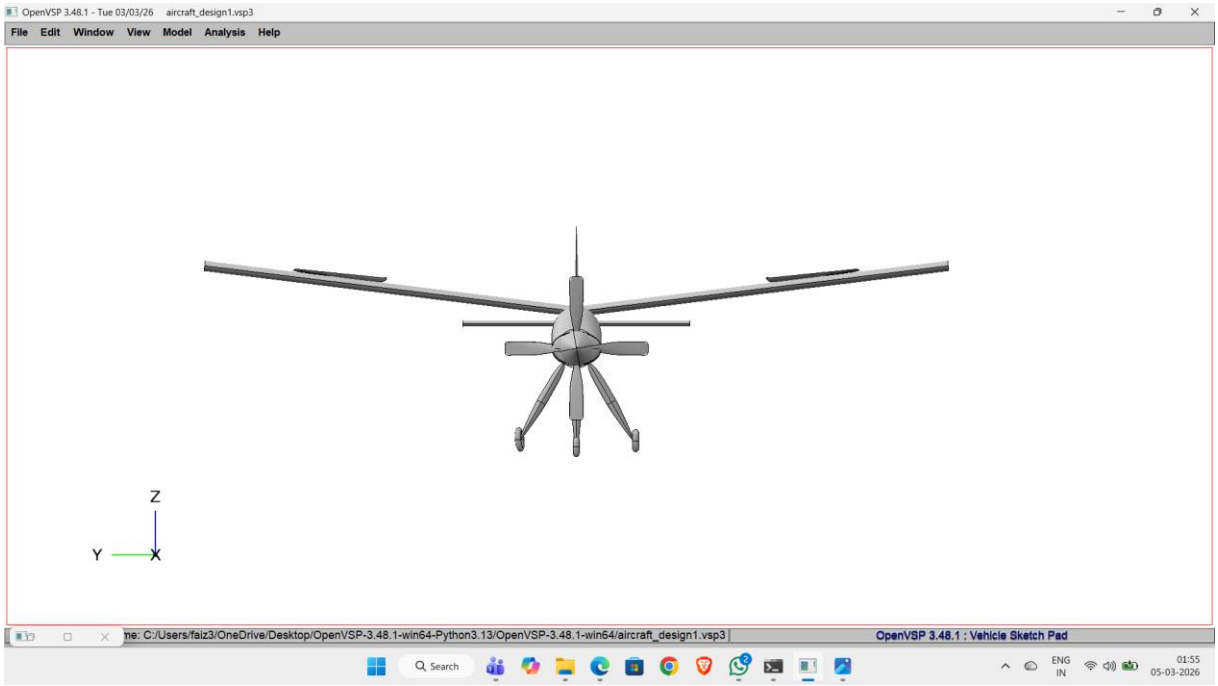

FIG 3 : Front View

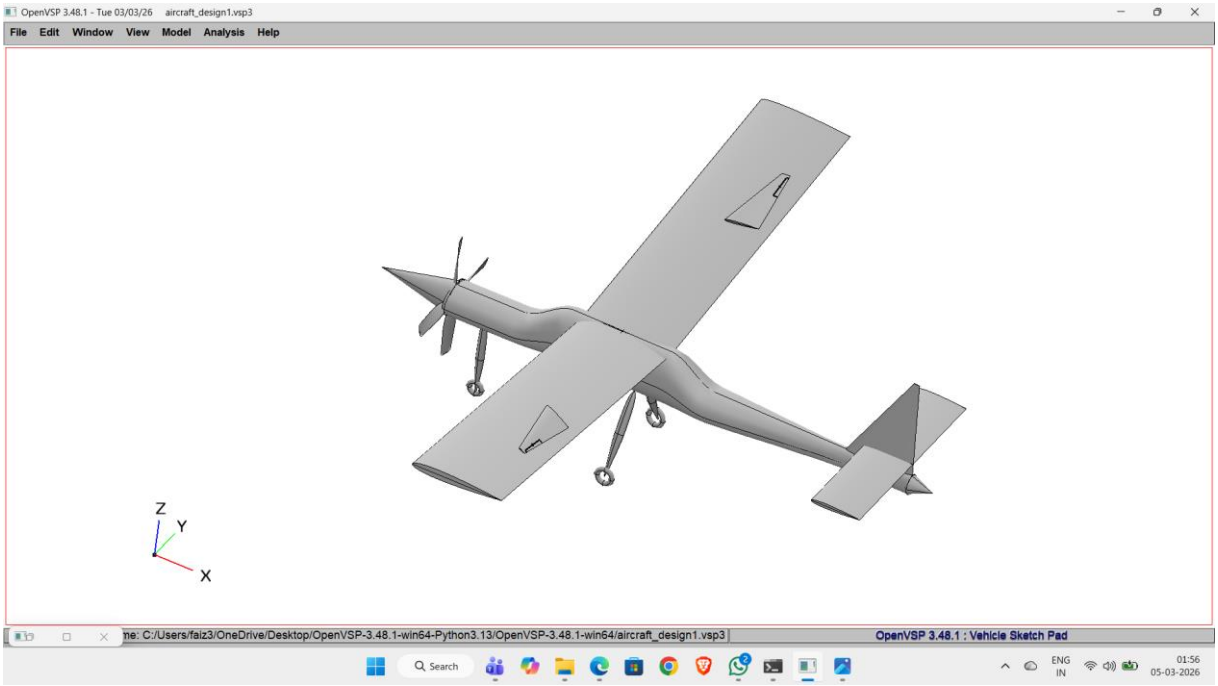

FIG 4 : Isometric (3D) View

These views demonstrate the arrangement of the fuselage, wings, tail structures, and propeller.

Conclusion

In this project, a simplified aircraft geometry inspired by the Cessna 210 configuration was successfully created using OpenVSP software. The fuselage, main wing, horizontal stabilizer, vertical stabilizer, and propeller were modeled and integrated to form a complete aircraft configuration.

The project provided valuable experience in parametric aircraft modeling and conceptual design techniques. It also highlighted the usefulness of OpenVSP as a powerful tool for visualizing aircraft configurations during the conceptual design phase.

The knowledge gained from this project can be applied to more advanced aircraft design studies including aerodynamic analysis and performance evaluation.